Draft version February 24, 2025
Typeset using LaTeX twocolumn style in AASTeX631

arXiv:2502.15028v1 [astro-ph.HE] 20 Feb 2025

A New Framework for AGN Accretion and Jet Feedback in Numerical Simulations

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ABSTRACT

Accurate modeling of active galactic nucleus (AGN) feedback, especially due to relativistic jets, is crucial for understanding the cool-core problem in galaxy clusters. We present a new subgrid method to model accretion onto and feedback from AGN in hydrodynamical simulations of galaxy clusters. Instead of applying the traditional Bondi formalism, we use a sink particle algorithm in which the accretion flux is measured directly through a control surface. A weighting kernel is used to reset the gas properties within the accretion radius at the end of each timestep. We implement feedback in the form of bipolar jets whose properties are tied to the accretion rate. The method is tested with a spherically symmetric Bondi gas flow problem and a Bondi-Hoyle-Lyttleton wind problem, with and without jet feedback. We discuss the reliability of this model by comparing our jet simulations with those in the literature, and we examine the dependence of test results on parameters such as the resolution and size of the jet injection region. We find that the sink particle model can account for the α factor in accretion measurement, and the accretion radius must be resolved with at least two zones to produce realistic black hole accretion. We also show how under-resolving the AGN feedback region in simulations can impact the feedback energy deposited and the jet dynamics. The code described here is the framework for a feedback model, described in a companion paper, that will use accretion disk modeling to more self-consistently determine the feedback efficiency.

Keywords: Active galactic nuclei (16), Hydrodynamics (1963), Astronomical simulations (1857), Galaxy clusters (584)

1. INTRODUCTION

A number of physical processes shape the observational properties of the hot, X-ray emitting intracluster medium (ICM) in galaxy clusters. These include radiative cooling, cosmic rays, thermal conduction, viscosity, turbulence, and black hole jet heating. Observations have shown that the cold gas formation rate at the centers of some galaxy clusters is much lower than expected based on X-ray luminosity, a discrepancy called the "cool-core problem" (Peterson et al. 2003). Cool-core clusters have high central densities and low central temperatures, corresponding to short cooling times, but generally exhibit $\sim 1\%$ of the star formation rate expected under these conditions (McDonald et al. 2018). Active galactic nuclei (AGN) are thought to be the major feedback source that prevents hot gas from cooling and reduces star formation (McNamara & Nulsen 2007; Harrison & Ramos Almeida 2024).

AGNs are among the most energetic objects in the Universe, characterized by extremely luminous cores powered by the infall of gas from a relativistic accretion disk onto a central supermassive black hole (SMBH). Two major feedback modes have been identified for AGN from observations (Fabian 2012). The radiative (or quasar/wind) mode operates when the SMBH accretes at a rate close to the Eddington limit; the mechanical (or radio) mode, characterized by radio-emitting jets, operates at lower accretion rates. Mechanical feedback is generally thought to heat the ICM more efficiently, and synchrotron emission from relativistic jets has been observed at radio frequencies for bright X-ray sources (Kolokythas et al. 2018). The estimated jet power of AGN-inflated cavities is correlated with the X-ray luminosity within the cluster cores (Bîrzan et al.

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2008). The X-ray cavities also show correlation with radio emission; an example is the NGC 1275 in the Perseus cluster. The Phoenix cluster is a rare contrasting example in which AGN feedback does not appear to fully offset gas cooling (McDonald et al. 2019).

In addition to X-ray cavities and radio bubbles, optical, infrared, and submillimeter observations have revealed molecular gas filaments in a number of cool-core clusters. In Hα and sometimes CO, observers have discovered radially extended cold filaments around the X-ray cavities (Conselice et al. 2001; Simionescu et al. 2018; Vantyghem et al. 2018; Olivares et al. 2019). Optical spectroscopy has shown that these filaments are ionized by shocks (Boselli et al. 2019). ALMA observations show that the filaments are entrained by the X-ray-emitting hot ICM, flowing inward toward the AGN and then rising alongside AGN jets and behind the bubbles they inflate (McNamara et al. 2016; Tremblay et al. 2018). In addition, larger-scale structures can be present, such as merger shocks and cold fronts, turbulence, and ripples produced by sound waves (Fabian et al. 2017).

Multiphase hydrodynamic simulations of galaxy clusters can now produce realistic feedback loops and reproduce some of the structures observed in cool-core clusters (e.g., Gaspari et al. 2017; Li et al. 2017; Ehlert et al. 2023), and there has been significant progress in understanding different heating mechanisms and how jet feedback is coupled with them (e.g., Yang & Reynolds 2016a; Martizzi et al. 2019; Bourne & Yang 2023). However, due to the large range of spatial and temporal scales involved in the problem, it is infeasible to resolve length scales down to the size of the black hole for a cluster-scale simulation. As a result, accretion and feedback are usually included through a phenomenological subgrid prescription. As an example, in adaptive mesh refinement (AMR) simulations, the accretion rate onto the black hole is usually determined by assuming that the subgrid flow is described by the spherically symmetric Bondi accretion model (Bondi 1952), with an arbitrary boost factor applied to the rate computed from densities and sound speeds measured on resolved scales, which may be considerably larger than the accretion radius (Booth & Schaye 2009). The energy input rate of the feedback is taken to be a constant fraction of this accretion rate, and an accretion rate-dependent switch decides between quasar-mode and mechanical-mode feedback (e.g., Richardson et al. 2016; Meece et al. 2017). In cosmological simulations, such as the EAGLE (Schaller et al. 2015), Horizon-AGN (Dubois et al. 2016), IllustrisTNG and TNG-Cluster (Nelson et al. 2019, 2024), Simba (Davé et al. 2019), and Obsidian (Rennehan et al. 2024) simulations, where resolution is on the scale of kiloparsecs, such approximations are frequently used. In some cases, to account for cold and hot gas, the boost factor can vary for gas of different temperatures (e.g., Steinborn et al. 2015). The feedback is mostly imposed through isotropic energy injection into the cells surrounding the SMBH.

In all these simulations, ad hoc parameters are used to follow a non-self-consistent black hole accretion history. The assumption of spherical accretion is unrealistic, and the necessary boost factor is resolution-dependent. For example, Negri & Volonteri (2017) have found that the approximate Bondi formula can lead to both over- and underestimation of BH growth, depending on resolution and on how variables entering the accretion rate equation are calculated. However, for a problem involving accretion and feedback, often no single separation scale exists; as resolution improves, the necessary subgrid physics varies. To accommodate these facts, many alternative accretion models have been investigated. For example, some groups have used different inputs to a Bondi-like equation, such as the large-scale properties of host galaxies (Anglés-Alcázar et al. 2015). Others take into account some disk structure or angular momentum effects, but without aiming to actually follow the accretion disks themselves (Power et al. 2011). Studies of convergence and validation for these models are still limited.

A major limitation of these "hot Bondi"-like accretion models is that they make no provision for thermal instability in the gas on scales between the resolution length element and the size of the accretion disk. Despite the fact that heating dramatically suppresses star formation in cool-core clusters, the Hα filaments strongly indicate the presence of cool gas. McCourt et al. (2012), by studying an ideal gravitationally stratified plasma, found that the non-linear saturation of thermal instability is governed by the ratio of thermal instability timescale $t_{\rm TI}$ to the dynamical ("free-fall") timescale $t_{\rm ff}$. When $t_{\rm TI}/t_{\rm ff} \leq 1$, the cold gas condenses, and the plasma develops an extended multiphase structure. For spherical systems, this criterion becomes $t_{\rm TI}/t_{\rm ff} \leq 10$. Further studies have confirmed that, when the ratio falls below this value, gas condensation occurs, and within a cluster potential it also precipitates into the region closer to the SMBH and is accreted by it (Tremblay et al. 2016; Voit et al. 2017; Nobels et al. 2022). The condensed clouds then cool and form a filamentary nebula structure, which can explain the Hα filaments. This scenario has come to be known as "chaotic cold accretion" (CCA; Gaspari et al. 2013). This discovery has motivated a way to treat accretion as an inflow through

a spherical region. For example, Gaspari et al. (2013) evacuated the gas within an accretion sphere in their simulations and used the mass of gas lost per timestep as the accretion rate for that step. Given that the flow becomes mostly supersonic at parsec scales, a complete evacuation is justified. However, given the turbulent and multiphase ICM structure involved, the exact kinematics of the sink-region gas is more complicated and involves both inflow and outflow. Methods that average density over a few large zones surrounding the SMBH do not consider these effects, and are thus inaccurate compared to a flux-based method that measures the actual mass inflow.

The notion of "sink particles" was first used in star formation studies. They were developed with the general aim of coping with regions of a flow that accrete infalling material but whose internal structure is unresolved. Sink particles were introduced by smoothed particle hydrodynamics (SPH) codes (Bate et al. 1995) as absorbers that accrete other particles that approach within a certain distance and meet various other criteria. Krumholz et al. (2004) were the first to introduce sink particles in the Eulerian, grid-based code ORION, built upon the AMR technique. In addition to determining the accretion rate, the gas mass accreted in a timestep, $\dot{M}\Delta t$, is also removed from the same region along with the related momentum and total energy. Later Federrath et al. (2010) introduced a sink particle implementation into the FLASH code (Fryxell et al. 2000; Dubey et al. 2008) with an approach that deviates considerably from the original (Krumholz et al. 2004) method; they tested whether gas is gravitationally bound to the accretor before adding it to the accretor's mass. Since then a number of implementations have been developed in various other grid-based codes, such as Enzo (Wang et al. 2010), RAMSES (Bleuler & Teyssier 2014), and Athena (Gong & Ostriker 2013). Most of these works incorporate various methods to calculate the mass flux and a further correction of the accretion region to smooth out numerical effects. These sinks provide a realistic way to model accretion without damaging the rest of the calculation on the grid.

In the absence of a detailed understanding of jet and wind launching from accretion disks, the methods for introducing feedback linked to accretion have typically also relied on adjustable parameters. The quasar-/radio-mode switch is one; others include the energetic feedback efficiency and the mass-loading factor. However, relativistic jet launching and propagation have been studied via (magneto)hydrodynamical simulation for decades (see the review by Martí 2019). Most recently, general relativistic magnetohydrodynamics (GRMHD) simulations have been able to generate jets self-consistently from rotating magnetized accretion flows (McKinney & Blandford 2009). These successes enable us to consider unifying simulation results on accretion disk scales with larger scales associated with thermal instability in cluster cores (Gaspari & Sądowski 2017). Effects of heating and feedback from relativistic jets on kiloparsec scales have been studied by Mukherjee et al. (2018) and Perucho et al. (2019), and updated physics with accretion disk structure and disk winds has been considered (e.g., Yuan et al. 2018; Costa et al. 2020). Recent efforts including Talbot et al. (2021), Huško & Lacey (2023), and Rennehan et al. (2024), incorporate varying accretion disk configurations into a unified model.

Several major questions concerning the details of AGN feedback and its interplay with the ICM have so far remained unanswered. In particular, how does the energy in AGN jets couple to the ICM? Why do some clusters have cool cores while others do not? What processes govern transitions between the two states? How do we explain objects such as the Phoenix cluster? A better understanding of these questions requires more knowledge of the plasma physics governing diffusive processes in the ICM (e.g., Yang & Reynolds 2016b). However, to address them, a more physically motivated subgrid model for AGN accretion and feedback is also required.

Motivated by these unresolved questions and previous models, we have developed a new subgrid framework for accretion and jet feedback in hydrodynamic simulations of AGN feedback using FLASH. The accretion model resembles a sink particle algorithm, measuring accretion onto the SMBH via numerical fluxes across a spherical surface instead of relying on the Bondi solution. With this formulation we further investigate different ways to inject jets and characterize mechanical feedback, and we compare the jet dynamics and feedback properties with other smaller-scale simulations to test the convergence of our model. Upon this framework, we are also building a feedback model that incorporates the evolution of the AGN accretion disk, to be described in a companion paper (Khan et al. 2024, in preparation, hereafter Paper II).

The paper is organized as follows. In § 2 we introduce the details of our accretion method and discuss convergence tests in the absence of feedback. In § 3 we describe the jet feedback model and examine the effects of different parameters on feedback. In § 4 we briefly survey the literature on jets in numerical simulations at different scales and compare our test results with previous work. We offer conclusions in § 5.